PLUG POWER INC PLUG S

October 01, 2026 - 10:10am EST
by katana

			2026	2027
Price:	1.94	EPS	-0.41	-0.16
Shares Out. (in M):	1,400	P/E	-5	-12.5
Market Cap (in $M):	2,716	P/FCF	-4	-4
Net Debt (in $M):	660	EBIT	-294	-171
TEV (in $M):	3,376	TEV/EBIT	-12	-20
Borrow Cost:	General Collateral			

Description

PLUG is one of the all-time most “successful” (as in long-lived, non-microcap) shitco stock promotes. It is one of my all-time favorite short positions, truly the gift that keeps on giving to short-sellers. I have been short continuously for 3 years and have booked something north of 100% profits as a percentage of my average position size. With the stock price currently near a 52-week low, half of its May peak price, and reasonably likely to double whenever the next pump of all speculative stocks happens, now is not the *best* time to enter a PLUG short. But there is never a truly bad time. You can do the analytical work, take a starter position to force yourself to pay attention, and then wait. The occasional little pumps, and even the big pumps, are opportunities. You just need to scale your share count up and down as the stock price moves up and down, which it will inevitably do. That core PLUG position you don’t cover is your perpetual funding short; the additions are a great trading vehicle that generates realized profits over and over.

Plug Power went public in 1999 as a pure-play provider of hydrogen power equipment, power generation plants, and related services. That sentence alone is, if you know anything about hydrogen power, enough to explain why PLUG is such a dog of a business. Hydrogen is the bottom of the barrel for both power sources and the industries that supply them. Although hydrogen is a “fuel” in the sense that you can burn it to produce thermal energy, in economic terms it’s effectively a battery, a way to get energy from one place where it can be efficiently generated to another place that needs energy but can’t efficiently be served by electricity. And it’s a very bad battery:

- Producing hydrogen requires expending far more energy to separate elemental hydrogen from more complex molecules than you get from burning the resulting hydrogen.
- Liquifying gaseous hydrogen requires cooling it to near absolute zero and keeping it there. The cooling and compression wastes 30-40% of its initial energy right from the start.
- Storing or transporting the compressed hydrogen requires heavy reinforced tanks and pipes, plus chillers at the end-point fueling stations, which adds further to the cost and wastes another ~15% of its initial energy.
- Even when compressed, hydrogen has by far the lowest volumetric energy density (units of energy per unit of volume) of any fuel. It is less than 1/6 as dense as gasoline and a bit over 1/3 as dense as LNG. That makes storing and transporting it even more expensive, per energy-value delivered.
- Hydrogen infiltrates and weakens metal containers over time and requires more expensive maintenance and replacement cycles.

In addition, hydrogen is only “green” (substantially reducing greenhouse gasses and global warming vs. alternatives) if it is produced through water electrolysis using green electricity sources. Only 1% of hydrogen produced in 2026 is green. 95% is “grey” hydrogen produced from natural gas or coal, and 4% is “blue” produced the same way but with carbon capture. Electrolysis costs multiples more than the grey methods.

This combination of factors drives the cost of green hydrogen, many years after people have been working on the problem and moderately lowering its cost, to multiples of traditional fuel costs. Natural gas currently sells wholesale in the U.S. for $3-4 per MMBtu. Storage, delivery, and intermediary costs bring the retail price up to $5-10 per MMBtu. Hydrogen produced via electrolysis costs $4.50-$6 per kg to produce, plus $2-5 per kg for storage and delivery, $6.50-11.00 total. It takes 7.5 kg of hydrogen to get 1 MMBtu of energy, which brings the delivered cost to $50-85 per MMbtu – roughly 10x that of U.S. natural gas.

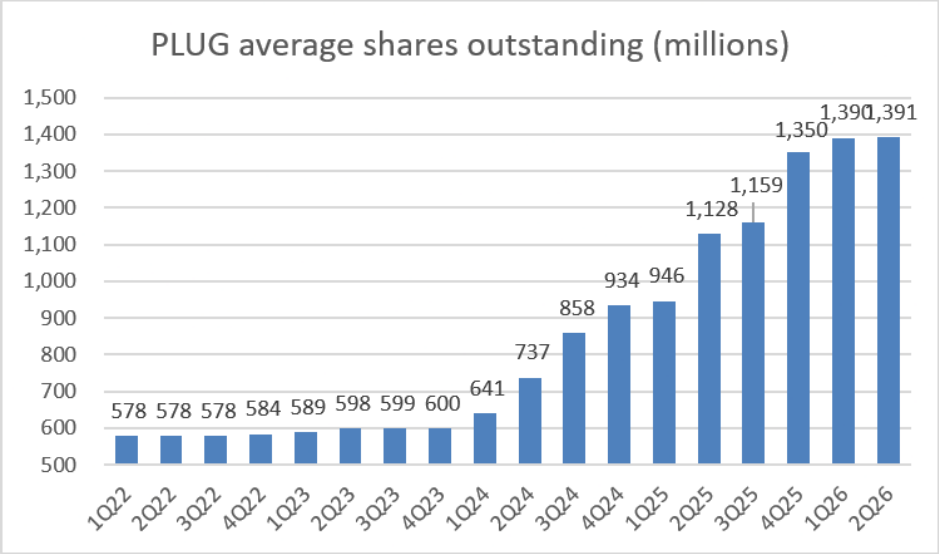
When looking at all the green-energy alternatives, there is no reason why anyone would choose hydrogen, except for one thing: There are significant “hard to abate” economic activities that can’t be electrified – long-haul transportation, shipping, and some industrial sectors. For them, there is no green alternative to fossil fuels apart from hydrogen. Net-zero advocates and policymakers, determined to set overall targets of literal net-zero rather than net-much-lower-than-now, have looked at these economic sectors and simply hand-waved assumptions that advancements in hydrogen engineering and cost reductions in green-energy electricity for electrolysis would bring hydrogen costs down into acceptable ranges. Those assumptions were unrealistic when made and have remained unrealistic after decades of trying and further evidence. Recent balanced attempts to estimate future cost reductions (Oxford Institute for Energy Studies; BloombergNEF; Harvard Belfer Center) forecast delivered costs of $5-9 per kg 5 years from now, $4-6 in 10 years, and $2.50-4.50 in 25 years – still multiples of existing alternative fuels. Here is a link to my favorite article on the topic, which, while older (2023) than the other three I mentioned, came up with the same cost forecast range for 2050 and is explicitly written to explain why more optimistic estimates are unrealistic.

Politicians still don’t care about realism; they still need their hydrogen plug figure (rim shot!) to pretend that net-zero is within reach, so they keep hydrogen in their net-zero plans and powerpoint slides and keep passing hydrogen subsidies or limited mandates. But at least in most countries, they are unwilling, as they should be, to do so as intensively as would be necessary to make hydrogen viable for many to use or almost anyone to produce.

That non-viability includes PLUG. The company went public in 1999. This year will be PLUG's 27th consecutive year of reporting operating losses, usually massive ones relative to its business size. The entire way, its management has made highly negative-NPV capital investment and contracting decisions, has

generally mismanaged the business, and has consistently overpromised and underdelivered. (They don't give traditional guidance, but they sometimes give somewhat vague and ad-hoc targets, which they usually miss.) Just to give a flavor: In the last 3 years, the company's gross margins were -57%, -99%, and -34%, and EBIT margins were -151%, -321%, and -207%.

PLUG has funded all these losses by almost continuously selling epic, probably record-setting amounts of stock. PLUG had 40 million shares outstanding after its IPO. It now has 1,400 million, 35x higher. Here is a chart showing the share count progression since 2022.



In 2025-2026 PLUG finally hit enough of a financial wall to force even this management team to scale back its dreams, cut costs, abandon the worst of its projects, sell off some assets to generate cash, and get out from under some terrible power sales agreements with pricing below cost. That has allowed management to give a target of "breakeven adjusted EBITDA" in 4Q. The "breakeven" word is not very meaningful to the business's future. The primary adjustments to adjusted EBITDA are not ones any self-respecting VIC member would credit. The capex and D&A are large and recurring, as you might expect for a heavy-industry business. Operating losses were still -174m (-51% margin, 6% of market cap) in 1H26 and are expected to be -294m (-36% / 11%) for full-year 2026 and -171m (-18% / 6%) in 2027. Moving in the correct direction, but still bad. Cash burn in 1H26 was -289m, far worse than operating losses. The asset sales will slow the cash burn and share sales only temporarily. More share sales are coming as soon as 4Q, if PLUG does not close the asset sale of a New York project that has been delayed by regulatory reviews.

Lower losses are coming at the cost of lower ambitions and growth prospects. PLUG's revenue backlog is at multi-year lows, down 40% in 2 years. Although less negative-NPV projects should be good for the business, that is not necessarily good for the stock. PLUG's revenue growth and management's promotion of coming new business lines and projects were historically two of the main things that excited retail day-traders. It is possible that management could eventually carve the business down to something that makes money, but, first, PLUG would then not be a good "story stock" to pump, and second, it still probably wouldn't even be much profits, especially per share. The profits will be spread among the 1.4 billion existing shares, plus whatever additional shares the company needs to sell between now and then.

The only real upside risk I see for this business is if a future administration and Congress go really crazy with subsidies. That seems unlikely. The Biden administration's $trillions in green-energy dollars from the Inflation Reduction Act were about as crazy as it is likely to get, in terms of extra deficit spending and indiscriminate green-energy spending, and even the IRA ended up being not all that crazy when it came to hydrogen. (Investors' hope for higher hydrogen subsidies is partly what drove the stock prices so crazy, and the actual regulations' dashing of those hopes is partly what drove the stock crash.) The Biden administration's regulations implementing the IRA money offered tiered subsidies of $3, $1, $0.75, and $0.60 per kilogram of hydrogen produced, depending on how much "lifecycle greenhouse gas" is associated with a particular batch of hydrogen. The killer for PLUG's subsidy dreams was that, to claim that your hydrogen meets the highest tier, you can't just buy electricity off the grid to produce it and point to some renewable electricity production that's theoretically available. Your plant's hydrogen must meet the requirements of

1. Additionality - the renewable energy source must have been added to the grid less than 36 months before your hydrogen plant enters service
2. Deliverability - the generation must be located in the same geographic grid region as your hydrogen plant
3. Temporal matching - Your plant must produce the hydrogen during the same hour that the renewable source is producing electricity. No credit for solar when it's dark outside, no credit for wind when it's still. This effectively prevents any hydrogen plant from economically running anything close to 24/7.

These existing IRA subsidies apply only for plants that begin construction by 2027. If a new administration returns with new subsidies, they will likely not be more generous than the Biden IRA ones were.

One other theoretical longer-term risk is that Western Europe, with its much higher – dare I say self-immolating? – commitment to net-zero goals and green-energy mandates/subsidies, could save PLUG. (A few other countries behave similarly but don't collectively move the needle; easiest to just lump them into "Western Europe.") Not only are the green-energy subsidies sometimes higher; thanks to its revulsion to producing its own fossil fuels, Western Europe pays 2-5x as much for shipments of liquid natural gas (LNG) as U.S. customers pay for their gas. Thus the LNG price is closer to hydrogen's price and makes

hydrogen “only” 3-5x more expensive, compared to 10x in the U.S. However, Europe has long had plenty of entrenched domestic businesses competing for those hydrogen mandates and subsidies. PLUG has pursued the European opportunity for years, and it has failed to make any real headway. Only 12-14% of PLUG’s recent revenues come from outside the U.S. As a potential small bonus short thesis, my macro/political best guess is that the European governments are, within a few years, going to be forced by either fiscal realities or their voters to give up or substantially scale back their lowest cost/benefit “green”-energy pursuits, which should include hydrogen. (I put "green" in quotation marks because most of Europe's self-described "green" energy comes from burning wood pellets.)

That’s really it. PLUG is business that may lose money forever and will at best make not-much after years more of losses. It’s in an inherently terrible industry that has little blue-sky upside to improve its outlook sufficiently through either technological advances or more favorable government policy. Stock dilution will continue to fund the losses. And an unsophisticated and/or cynical shareholder base will keep jerking the stock price around so you can scale-trade it profitably until you retire. These are a few of my favorite things. Enjoy!

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Catalyst

- Ongoing losses
- Ongoing stock sales
- Temporarily higher stock prices are a feature, not a bug

Messages

Subject	Wow
Entry	10/01/2026 07:28 PM
Member	Djokovic1

How is this dumpster fire still alive...You are right... I haven't seen such atrocious financials sustain without bankruptcy for such a long period of time.